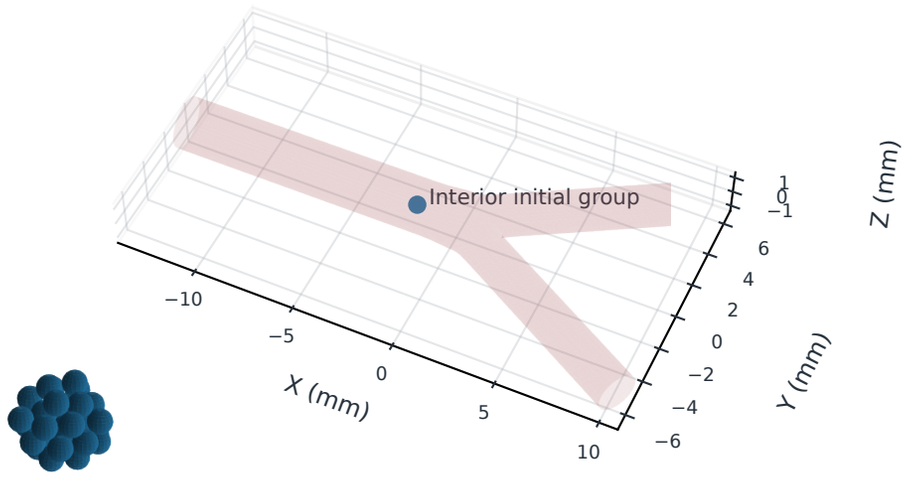
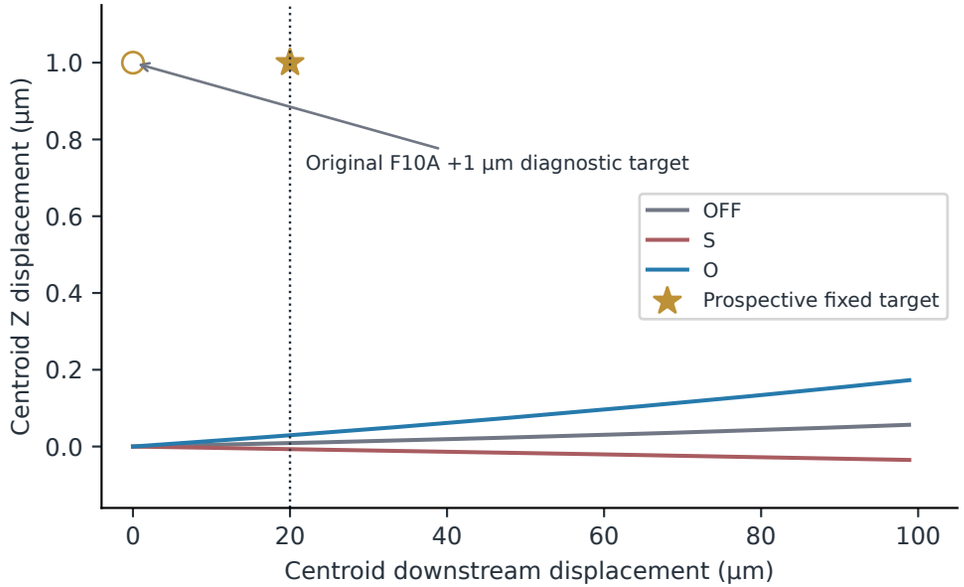
Interior-lumen navigation: measured control time and limited transverse authority

a Same original Y-vessel and CFD

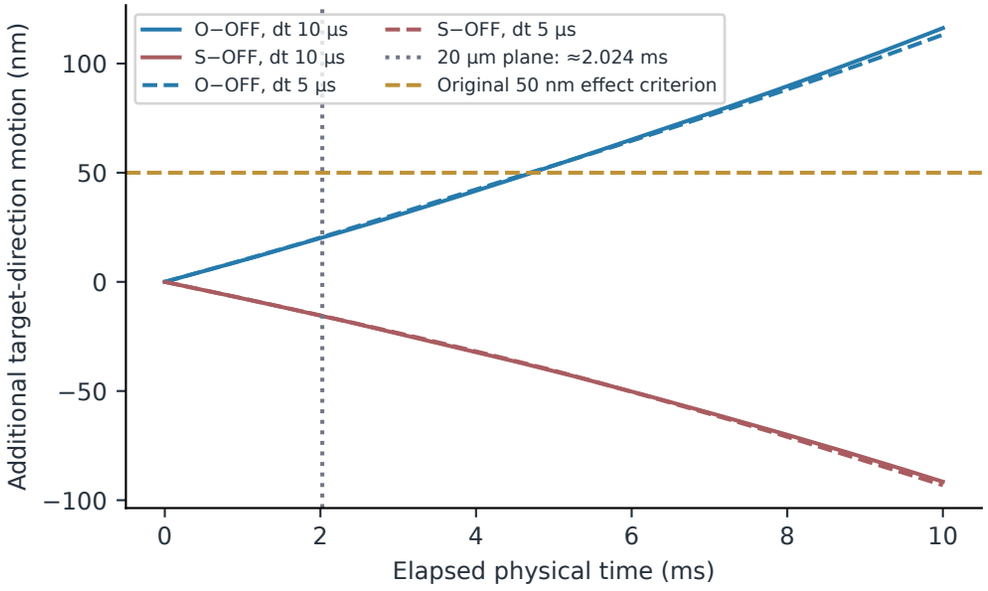


Initial 32; R=1.78 µm

b Actual saved centroid paths; unequal axis scales



c Paired effect from complete saved trajectories



d Control time depends on the target plane

At the 20 µm downstream plane

O−OFF: 20.32 / 20.53 nm (10/5 µs)
Paired step-size difference: 0.208 nm
Error to fixed +1 µm target: 0.971 µm

O points toward +Z, but misses the 50 nm effect criterion.
Field/CFD support observed throughout 10 ms;
actual field-exit time is right-censored.

Navigation to the proposed target: NOT ESTABLISHED
Near-wall F8 target is a separate historical case.

1 kPa is the initial local pressure median. Curves are actual F10A trajectories; plane crossings interpolate real 50 µs brackets.
The downstream target is a prospective protocol, not a completed historical navigation task. EXPLORATORY / training_ready=false.